

From Seed Papers to an Ontology of POTS

What we built, how the links were found, and where the ontology stands now

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Summary

We are building the literature layer for a computable phenotype of postural orthostatic tachycardia syndrome (POTS): a vocabulary of topics (possible causes, tests, conditions, treatments) and an ontology of links between them, each link backed by a passage from a seed paper. We started from seven seed papers. We wrote the questions the ontology must answer, built a 72-term vocabulary, and had an AI reader extract quoted claims, which gave a first ontology of 132 links. We then made the reading reproducible: scripted Claude readings, three local or hosted *System One* decision models that answer typed questions paragraph by paragraph, and a vote. The current ontology keeps a link only when Claude and at least one System One model find it in the same paper.

Seed papers	7	Vocabulary terms	72	Literature links now	120
Paragraphs read	236	Possible causes (mechanisms)	7	kept from the first 132	105
Readers	4	Relationship types	10	new	15
Claude readings per paper	3	Taxonomy (is_a) edges	102	Quoted passages behind them	265
Corpus articles harvested	2,668	Classes	38	Links for review	158

Table 1: Where the work stands. The links are on the review branch (pull requests #23–#26) and on the public site.

1 The path from seed papers to ontology

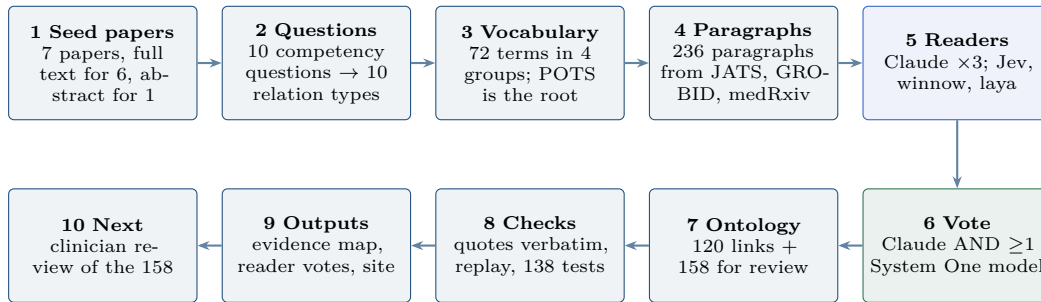


Figure 1: The whole path. Steps 1–3 fix what the ontology may say; steps 4–6 decide what the papers say; 7–9 are generated and checked.

Stage	What we did	Result
First build (1 Oct)	Wrote the competency questions; five AI reader agents (Claude) read the seven papers and copied 407 claims with quotes; new topics were merged or dropped; links were generated from the claims; PubMed corpus harvested and tagged	72 terms, 132 links, 401 passages; 2,668 articles
Reproducibility test (2 Oct)	Turned the reader into a script (claude -p) with fixed, hashed inputs and read every paper three times	86% link overlap between readings; 115 of 132 links recovered
Paragraphs	Split the papers into paragraphs with provenance: PMC JATS XML for 4, GROBID for the 2 PDFs, the medRxiv abstract	236 paragraphs; 350 of 362 text quotes found
System One readers	Asked decision models four typed questions per paragraph: winnow and laya on this Mac, Jev on TypeSafe's service (open-licence papers only)	Local models repeat byte for byte; Jev 93–96%
Vote and rebuild	Combined the readers: a link needs Claude and at least one System One model in the same paper	120 links, 158 for review
Publishing	Evidence map and a reader-votes page with every question and answer; public site updated	\$7

Table 2: What was done, in order.

2 Seed papers, questions and vocabulary

Key	Paper	Text used	Licence	Jev?	Paragraphs	Links now
lau2026	Lau et al., review, <i>Heart Lung Circ</i> 2026	PDF → GROBID	CC BY 4.0	yes	46	73
chung2026	Chung & Raj, review, <i>JAMA</i> 2026	PDF → GROBID	AMA, reserved	no	46	55
angeli2024	Angeli et al., <i>Sci Rep</i> 2024	PMC JATS	CC BY 4.0	yes	25	43
gibbons2013	Gibbons et al., <i>PLoS ONE</i> 2013	PMC JATS	CC BY 4.0	yes	42	33
low2009	Low et al., <i>J Cardiovasc Electrophysiol</i> 2009	PMC JATS	author manuscript	no	31	31
zhang2022	Zhang et al., <i>Cureus</i> 2022	PMC JATS	CC BY 3.0	yes	42	21
larsen2025	Larsen et al., long-COVID POTS, medRxiv	abstract	no CC licence	no	4	9
					236	265

Table 3: Seed papers. *Links now*: paper-and-link pairs in the current ontology (a link can rest on several papers). Jev, a hosted service that keeps requests, was sent only the openly licensed papers.

Competency question	Relation	Direction
Which possible causes are proposed for POTS?	proposed_mechanism_of	cause → POTS
Which tests give evidence for a cause, or for POTS?	evidence_for	test → cause
Which treatments, when prescribed, suggest the clinician suspected a cause?	indicates	treatment → cause
Which treatments are used for POTS or a condition?	treats	treatment → POTS
Which conditions co-occur with POTS or a cause?	associated_with	condition – POTS
Which events precede and trigger POTS or a cause?	trigger_of	event → POTS
What must be ruled out before POTS can be diagnosed?	differential_diagnosis_of	condition → POTS
What does a test measure? / Which tests make up a battery?	measures / part_of	test → quantity / battery
What kind of thing is each topic?	is_a	hand-kept taxonomy

Table 4: The ten questions, one relation each. Every literature link points toward POTS.

The vocabulary has 72 terms: 7 possible causes (hyperadrenergic, neuropathic, hypovolemic, deconditioning, autoimmune, post-infectious, cerebral hypoperfusion), 21 tests and measurements, 26 conditions, triggers and differential diagnoses, and 18 treatments, plus POTS itself and 9 quantities and test batteries. It was fixed before the readers below ran, so all of them chose from the same list.

3 The readers

Reader	Reads	Asked	Repeats exactly?	Cost / time
Claude (Opus 5)	whole paper, tables included, 3 times	“List every link, using only these terms, and copy the sentence that says it”	no; 86% of links match	Claude plan, +2% of weekly limit; 62 min
Jev 1.13 (TypeSafe)	one paragraph at a time	the four questions below	no; 93–96% match	\$0.49; minutes
winnow e4b (local)	one paragraph at a time	the four questions below	yes, byte for byte	free; 40 min
laya en (local)	one paragraph at a time	the four questions below	yes, byte for byte	free; 11 min

Table 5: The four readers. Jev, winnow and laya are *System One* models: they never write text, they return a probability for each answer option.

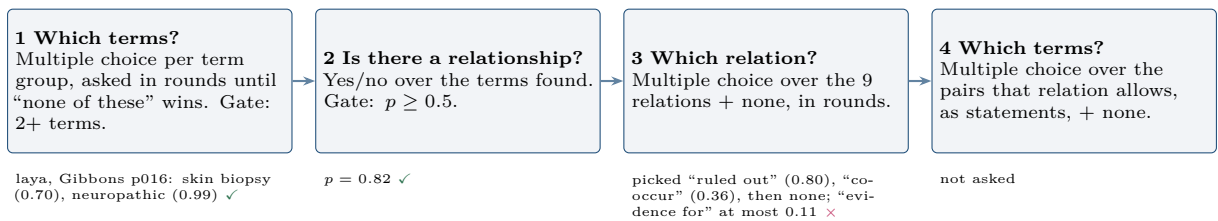


Figure 2: The four questions each System One model answers for every paragraph, with one real trace: why laya did not find “skin biopsy is evidence for a neuropathic cause” in Gibbons 2013.

How each reader did on its own

Measured against the first ontology (132 links), before any of these readers was used to build anything, so the comparison is not circular. *Recall*: share of those links the reader found. *Precision*: share of the reader’s links that were in the first ontology (a link not in it is not necessarily wrong).

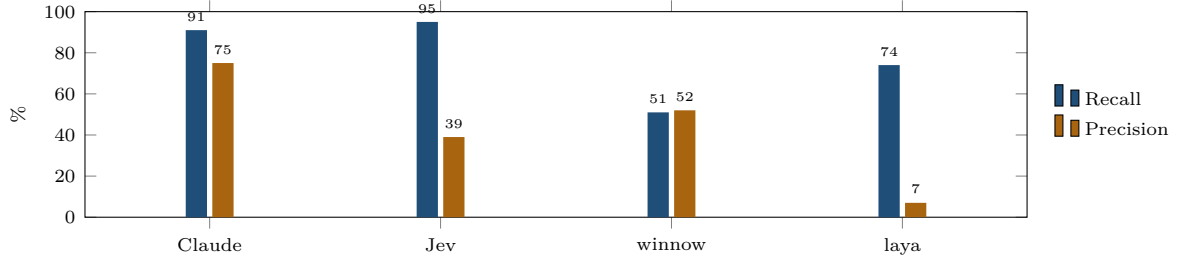


Figure 3: The four openly licensed papers, where all four readers ran (93 reference links). Links found: Claude 114, Jev 227, winnow 91, laya 958. Over all seven papers: Claude 87% / 80%, winnow 55% / 46%, laya 73% / 7%.

Jev matches Claude’s recall but finds many more links that the first ontology lacked. winnow is cautious. laya picks almost every option it is offered and cannot be used alone. No reader is good enough by itself, and the two kinds make different mistakes: the System One models cannot see tables, Claude sometimes misses a relation the text states plainly.

4 Combining the votes

For each paper and each candidate link we recorded which readers found it. The rule:

A link enters the ontology for a paper when **Claude** (2 of 3 readings, with a word-for-word quote) **and at least one System One model** find it in that paper. Other links that two or more readers agree on go to the **review list**. Links only one System One model found are ignored.

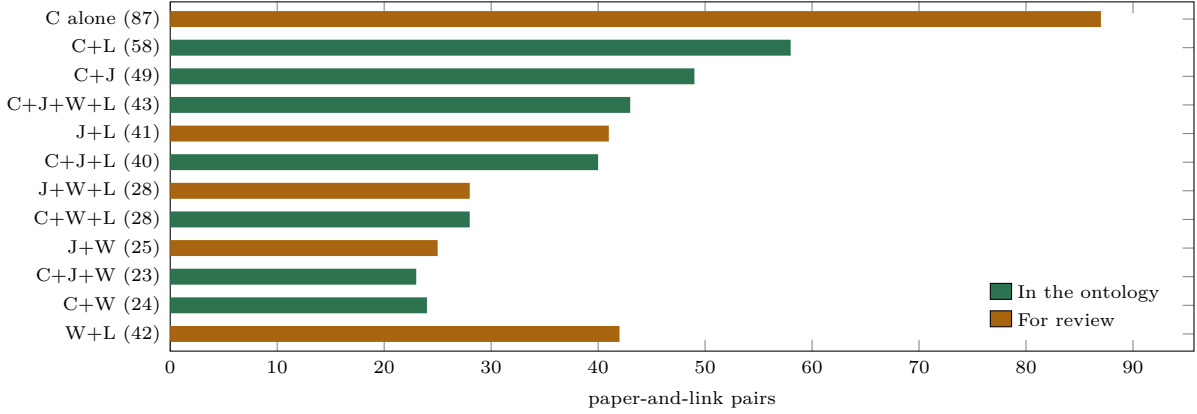


Figure 4: Which readers agreed. C = Claude, J = Jev, W = winnow, L = laya. 265 pairs met the rule; 223 went to review (158 distinct links). Not shown: single System One picks, ignored (laya 2,239; Jev 91; winnow 44).

5 The ontology now

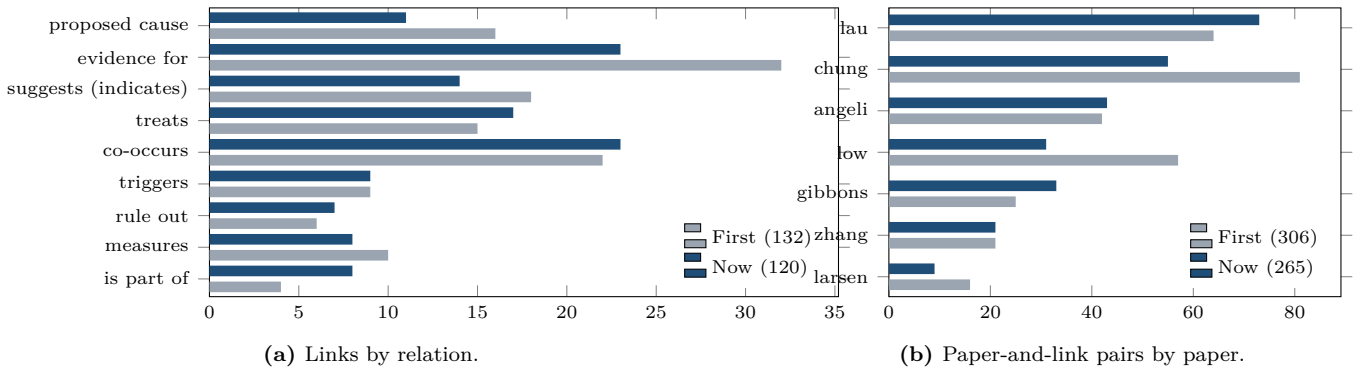


Figure 5: The first ontology against the current one. Lau, Gibbons and Angeli gain: Jev read them. Low 2009 and Chung lose most, because much of their evidence is in tables, which only Claude reads.

Change	<i>n</i>	Examples
New links	15	long COVID triggers the post-infectious cause; functional neurological disorder is ruled out; prokinetics and pyridostigmine treat GI dysmotility; tilt, active stand and cerebral blood flow are part of orthostatic challenge testing
Dropped: two readers never agreed	12	ambulatory monitoring links; hypermobility → neuropathic; post-infectious triggers autoimmune
Dropped: only Claude found it	8	blood volume test for hypovolemia; renin and aldosterone; skin biopsy for small fibre neuropathy (most rest on tables)
Dropped: only System One models found it	6	post-infectious → neuropathic; prokinetics suggest GI dysmotility
Dropped: Claude’s quote not word for word	1	phenobarbital (Low 2009: the quote joins two sentences that are not adjacent)

Table 6: What changed. Dropped links are on the review list, not deleted.

All seven possible causes keep their link to POTS, each backed by Claude and a System One model. Of the 120 links, 49 rest on one paper and 71 on two to six. Of the 265 passages behind them, 191 are marked *supports* and 74 *partial*; 131 were confirmed by one System One model, 91 by two and 43 by all three.

6 Reproducibility and checks

What	How it is pinned or checked
Inputs	Instructions, vocabulary, output schema and paper texts are hashed into every stored call; texts are not committed (copyright), their hashes are
Claude	Model, effort and Claude Code version recorded; three readings and a 2-of-3 rule because sampling cannot be fixed; a call answered by any other model is rejected (Opus 5.5’s biology safeguards had silently switched one call to Opus 5)
System One models	Model digests and Ollaya version recorded; winnow and laya repeat byte for byte (second run’s hashes stored); Jev pinned to jev-1.13.0 , its second run stored, 93–96% of links match
Paragraphs	GROBID 0.9.0 image pinned by digest; paragraph text identical across two GROBID runs; ids and text hashes in paragraphs.tsv
Assembly	assemble_ensemble.py -check rebuilds the excerpt file byte for byte from the stored readers, with no model calls
Quotes	every kept quote is found word for word in the paper text or its parsed paragraphs (351 of 351)
Tests	138 pass, including the replay, every link backed by both kinds of reader, and a removed link leaving the database

Table 7: What makes the result reproducible.

7 Outputs

Output	Where
The ontology	config/relations.yaml (120 links + 102 taxonomy edges), config/relation_excerpts.yaml (quotes, readers, review list)
Evidence map	every link with its quote and readers; public site /map/
Reader votes	every vote, six cases with the exact questions and answers; public site /votes/
Stored readers	extraction/2026-10-02/ (Claude), extraction/2026-10-02-decision/ (paragraphs, Jev, winnow, laya)
Corpus	data/articles.jsonl (2,668 articles) and derived tables
Public site	https://jeonglab-bcm.github.io/pots-phenotyping-site/ (JAMA text withheld; whole paragraphs only from CC BY papers)

Table 8: Where to find everything.

8 Limitations and next steps

- **No clinician has reviewed the links.** Agreement between readers measures consistency, not truth. The 158 links on the review list, and the 15 new links, are the place to start.
- **Tables are invisible to the System One models.** Links that rest only on a table (common in Low 2009 and Chung) cannot pass the rule. Adding table rows as their own blocks would fix this.
- **Jev read four of the seven papers**, because it keeps requests and three papers are not openly licensed. A local model at Jev’s level, or a zero-retention agreement, would let it read all seven.
- **The vocabulary is fixed.** Claude listed 114 topics it wanted but could not use, such as microvascular dysfunction, Chiari malformation and mitral valve prolapse. These are candidates for the next vocabulary review.
- **GROBID loses some text in the JAMA layout** (first letters of a few paragraphs), and figure text sometimes leaks into paragraphs.